

kira2diffeq Manual

Differential equations from Kira reduction output

Project version: local repository guide

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Contents

1. Project scope

`kira2diffeq` is a C++ utility that reads Kira-style reduction output and constructs differential equations for master integrals.

It performs these operations:

1. Read topology and kinematic configuration.
2. Parse `kira2math` reductions and `masters.final`.
3. Differentiate master integrals with respect to selected invariants.
4. Reduce shifted integrals back to masters.
5. Assemble and export DE matrices.

This package does not perform diagram generation or IBP solving.

2. Build and verification

2.1 Prerequisites

- CMake ≥ 3.16
- GiNaC
- yaml-cpp
- A C++17 compiler

2.2 Standard commands

```
cd /path/to/kira2diffEq
cmake -S . -B build
cmake --build build -j4
ctest --test-dir build --output-on-failure
```

2.3 Regression coverage

- `test_sunrise`: parses config, reduction, differentiation, matrix assembly.
- `cli_smoke`: end-to-end CLI run using local fixtures.

3. Command line usage

Executable: build/kira2diffeq-cli

3.1 Options

Usage: kira2diffeq-cli [options]

```
-c <dir>    Config directory (integralfamilies.yaml, kinematics.yaml)
-m <file>    kira2math .m output file
-f <file>    masters.final file
-t <name>    Topology name
-v <var>     Kinematic variable (default: all)
-o <fmt>     Output: text | math | all (default: text)
-e          Use epsilon (d = 4 - 2*eps) in output
-O <file>    Write output to file instead of stdout
-h          Show this help
```

3.2 Example

```
./build/kira2diffeq-cli \
-c tests/fixtures/cli_smoke \
-m tests/fixtures/cli_smoke/reduction.m \
-f tests/fixtures/cli_smoke/masters.final \
-t cli_smoke \
-v t \
-o text \
-O /tmp/cli_smoke_de.txt
```

Output file is created when -O is given.

3.3 Output formats

- **text**: default, matrix form with MI_i notation.
- **math**: Mathematica syntax blocks.
- **all**: all variables in Mathematica format.
- **-e**: substitutes $d \rightarrow 4 - 2 \text{ eps}$ in output.

3.4 Debug trace

differentiate_master is silent by default. Enable detailed logs with:

```
export KIRA2DIFFEQ_DEBUG=1
./build/kira2diffeq-cli ...
```

4. Required inputs

4.1 Configuration directory

Required files:

- `kinematics.yaml`
- `integralfamilies.yaml`

4.2 `kinematics.yaml`

Expected structure:

- incoming/outgoing momenta
- kinematic invariants
- scalarproduct rules
- optional momentum conservation
- optional `symbol_to_replace_by_one`

4.3 `integralfamilies.yaml`

Expected entries:

- `topology name`
- `loop_momenta`
- `top_level_sectors`
- `propagators`

Example snippets:

```
# standard propagator
- ["l1+p1-k2", t]

# bilinear ISP
- {bilinear: [{"k1", "l1"}, 0]}
```

4.4 `masters.final` and reduction file

`masters.final` has one line per master, for example:

```
cli_smoke[1]
```

`kira2math` lines map targets to linear sums of masters:

```
topo[1,2,0] -> +topo[0,1,0]*(1), +topo[1,1,0]*(-2)
```

5. Internal architecture

- **parser**: config and reduction data.
- **differentiator**: derivative rules + Gram-chain substitutions.
- **reducer**: map shifted integrals into master basis.
- **exporter**: text and Mathematica output.

Core files:

- `src/config_reader.cpp`
- `src/math_parser.cpp`
- `src/differentiator.cpp`
- `src/reducer.cpp`
- `src/exporter.cpp`
- `src/main.cpp`

6. Troubleshooting

6.1 Missing reduction entries

Warning pattern:

Warning: integral X[...] not found in reduction table

Interpretation: a shifted integral from differentiation is not reducible and not a master.

Fix list:

- add the missing reduction rule to `.m`
- check topology name/ordering and propagator definitions
- verify invariant symbols between config and reduction are identical

6.2 Parser errors

- check YAML structure in both config files
- ensure `kira2math` lines use the expected `target -> +master*(coeff)` form
- verify the selected topology with `-t`

7. Project-context notes

For a staged collinear workflow, keep regulated substitutions (for example p_{12}) in intermediate algebra and only apply strict limits after cancellations and checks.

8. Quick project map

- `MANUAL.md`
- `MANUAL.tex`
- `CMakeLists.txt`
- `tests/CMakeLists.txt`
- `tests/fixtures/cli_smoke/`
- `src/` and `include/`

9. Appendix

9.1 Smoke workflow

```
cmake --build build -j4
./build/kira2diffeq-cli \
  -c tests/fixtures/cli_smoke \
  -m tests/fixtures/cli_smoke/reduction.m \
  -f tests/fixtures/cli_smoke/masters.final \
  -t cli_smoke -v t -o text -O /tmp/cli_smoke_de.txt
```